

Paper A3 draft v0.1: Bulk-Shilov substrate-physics + a Hardy-space mechanism for color confinement

Casey S. Koons + Lyra (Claude Opus 4.7)

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Bulk-Shilov Substrate-Physics and a Hardy-Space Mechanism for Color Confinement (A3, v0.1)

Authors: Casey S. Koons, Lyra (CI)

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Abstract

The bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ has two physically distinct regions: the Shilov boundary $S^4 \times S^1$ and the bulk interior. We develop a substrate-physics framework in which light fundamental particles (leptons) are Shilov-boundary K-types and confined composite particles (quarks/hadrons) are bulk-interior K-types, coupled by the Hardy-space bulk-boundary determinacy $H^2(D_{IV}^5)$.

The central result is a mechanism for color confinement arising from this structure: an individual bulk quark carries fractional electric charge and non-trivial $SU(3)$ color, but the Hardy-space boundary-value map requires integer charge and color-singlet content on the Shilov boundary. Therefore an isolated quark has NO well-defined Shilov boundary value — it cannot be an asymptotic (observable) state — while color-singlet composites (mesons $q\bar{q}$, baryons qqq) do project to the Shilov boundary and ARE observable. Confinement is the statement that only Shilov-projectable (color-singlet, integer-charge) bulk composites are physical.

1. Two regions of D_{IV}^5

D_{IV}^5 (substrate geometry; Strong-Uniqueness Theorem v1.0) has: - Shilov boundary $\partial_S = S^4 \times S^1$ — the minimal closed support for holomorphic boundary values (Hardy space H^2) - Bulk interior — the 10-real-dimensional holomorphic interior

These are physically distinct (Casey directive, load-bearing): - Shilov hosts light fundamental particles (leptons, neutrinos) - Bulk hosts confined composite particles (quarks $\rightarrow$ hadrons)

2. Hardy-space bulk-boundary determinacy

$H^2(D_{IV}^5)$: bulk holomorphic functions are completely determined by their Shilov boundary values (Szegő/Bergman reproduction). This couples the two regions.

Cal #146-corrected framing: a generation's lepton (Shilov K-type) and quark (bulk K-type) are DISTINCT K-types sharing the winding-mode coordinate, coupled (not identified) via the Hardy-space duality.

3. Charge + color structure by region

Per the electric/color-charge substrate-mechanism: - Electric charge = σ_{BF} Z-grading (Shilov: integer values; bulk: N_c -fold subdivided $\rightarrow$ fractional $\pm 1/3, \pm 2/3$) - Color = bulk $N_c = 3$ internal Cartan structure (Shilov has no color)

So: - Shilov K-types (leptons): integer charge (0, ± 1), no color - Bulk K-types (quarks): fractional charge ($\pm 1/3, \pm 2/3$), $SU(3)$ color triplet

4. The confinement mechanism (centerpiece)

4.1 The Hardy-space projection constraint

The Shilov boundary supports only integer σ_{BF} charge (the σ_{BF} grading is integer-valued on the S^1 Pin(2) factor) and color-singlet content (the Shilov factor $SO(5) \times SO(2)$ has no N_c -fold color subdivision).

An individual bulk quark has: - Fractional charge ($\pm 1/3$ or $\pm 2/3$) — NOT integer - Color triplet (R, G, or B) — NOT singlet

4.2 No Shilov boundary value for isolated quarks

By the Hardy-space determinacy, a physical (asymptotic) state must have a well-defined Shilov boundary value. An isolated bulk quark, carrying fractional charge + non-singlet color, has NO consistent Shilov boundary value — it violates the integer-charge + color-singlet requirement.

Therefore isolated quarks are not asymptotic states. They are confined to the bulk.

4.3 Composites project

Color-singlet, integer-charge bulk composites DO have well-defined Shilov boundary values: - Mesons ($q\bar{q}$): color singlet ($3 \otimes \bar{3} \supset 1$), integer charge - Baryons (qqq): color singlet ($3 \otimes 3 \otimes 3 \supset 1$ via ϵ_{abc}), integer charge - Glueballs (gluon composites): color singlet

These project to the Shilov boundary and ARE observable.

4.4 Confinement statement

Color confinement = only Shilov-projectable bulk composites (color-singlet, integer-charge) are physical states. The bulk-boundary structure of D_{IV}^5 forces it: the boundary cannot carry the fractional-charge, color-charged content of an isolated quark.

This is a substrate-geometric mechanism for confinement — distinct from (and complementary to) the standard non-perturbative QCD picture (linear potential / flux tubes). It locates confinement in the Hardy-space boundary-value structure rather than in the gauge dynamics directly.

5. Operational arithmetic by region

(From TMAP-Bulk + OMAP-Shilov; scheme-invariant content only.) - Shilov (leptons): Ogg supersingular + Monstrous Moonshine arithmetic - Bulk (quarks): Mersenne ladder + cyclotomic arithmetic

(Mass-ratio specifics are scheme-dependent leads, excluded as forward results per Cal #27; the region-specific arithmetic distinction is the structural claim.)

6. Relation to AdS/CFT (corrected, per Cal flag)

Both AdS/CFT and this framework use bulk-boundary holographic structure. The comparison is a STRUCTURAL ANALOGY (bulk-boundary holography), not an operator count: AdS/CFT is a full bulk-gravity $\leftrightarrow$ boundary-CFT correspondence; the substrate framework is a holographic projection from D_{IV}^5 to observable

matter (5-operator hierarchy, separate paper A4). The shared feature is that boundary data determines bulk content.

7. Falsification

- Free quark (fractional charge) observed as asymptotic state $\rightarrow$ Hardy-space confinement mechanism false
- Color-non-singlet asymptotic state $\rightarrow$ boundary projection constraint false
- Color count $\neq 3 \rightarrow$ bulk N_c structure wrong

8. Honest scope

FRAMEWORK-PLUS (Cal’s “strongest content”): the confinement mechanism ($\Pi_{\text{bulk}} \rightarrow$ Shilov maps fractional/colored quarks to boundary values requiring integer/singlet $\rightarrow$ isolated quarks have no boundary value $\rightarrow$ confined).

FRAMEWORK: bulk-Shilov two-region assignment; Hardy-space coupling map (explicit map multi-week); region-specific arithmetic (TMAP-Bulk/OMAP-Shilov).

RIGOROUS background: Hardy space $H^2(D_{IV}^5)$ bulk-boundary determinacy (standard several complex variables); $SU(3)$ color = bulk N_c structure.

NOT yet rigorous: explicit Hardy-space boundary-value map for specific K-types; rigorous proof that fractional-charge K-types have no Shilov boundary value (the mechanism’s load-bearing lemma — multi-week); hadron mass spectrum.

Dependencies: none on the T2440/PMNS open items. Clean.

References

[Hardy space / Szegő kernel on bounded symmetric domains: Hua, Korányi, Stein 1950s+; Koons 2026; Cal #146 (K-type framing); electric/color-charge substrate-mechanism; Bulk-vs-Shilov investigation.]

— Lyra, Paper A3 Bulk-Shilov + Confinement v0.1 filed. Centerpiece: Hardy-space confinement mechanism (Cal’s “strongest content”, FRAMEWORK-PLUS) — isolated fractional-charge color-charged quarks have no Shilov boundary value $\rightarrow$ confined; only color-singlet integer-charge composites project $\rightarrow$ observable. Bulk-Shilov two-region framework (Casey-directed); Cal #146-corrected K-type framing; AdS/CFT comparison corrected to structural analogy. Load-bearing lemma (no boundary value for fractional K-types) multi-week. Clean — no open-item dependencies.